Taehyung Noh

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Research Interests

My research interest lies in Human-AI Interaction, with a focus on developing trustworthy and personalized AI systems.

Experience

- Hanyang Human-Centered Computing Laboratory [ ] — *Jan. 2022 - Present*
  Research Assistant — Seoul, Korea
- Ajou Human-Computer-Interaction Lab at Dept. of Software — *Aug. 2020 - Dec. 2021*
  Undergraduate Research Assistant — Suwon, Korea

Education

- Hanyang University — *Mar. 2022 - Present*
  M.S. & Ph.D. Integrated Student, Department of Artificial Intelligence (Advisor: Kyungsik Han) — Seoul, Korea
- Ajou University — *Mar. 2018 - Feb. 2022*
  Bachelor of Engineering, Department of Software — Suwon, Korea

Projects

- Industrial AI Data Preprocessing Platform
  Present
  - Developing an AI data preprocessing platform for industrial applications.
  - Providing an integrated solution for efficiently processing and analyzing data across various industrial domains.
- PALETTE: Value-Based User Understanding
  First Author — 2026
  - PALETTE is a theory-driven benchmark, grounded in Schwartz's Theory of Basic Human Values, for evaluating whether LLMs can infer a user's latent values from behavioral cues and transfer them to new contexts.
  - Constructed 25,200 dialogues across 400 personas spanning two tasks (Dialogue-Based Understanding and New-Context Application), revealing that LLMs generate appropriate responses without truly comprehending the underlying values.
- "Can LLMs Persuade Humans with Deception?" (CHI 2026)
  Third Author — 2026
  - Developed a taxonomy of LLM deceptive strategies and conducted a large-scale empirical study on whether LLMs can persuade humans through deception.
  - Investigated how different deceptive strategies affect human decision-making in human-AI interaction.
- TRIPLE (AAAI)
  First Author — 2025
  - TRIPLE (Theory-guided Reasoning for Intent and habIt Profiling with LLMs for pErsonalization) is a novel framework that integrates dual-process theory and the Theory of Planned Behavior (TPB) into LLM-based user modeling.
  - Constructs both habitual and intentional behavior profiles, then generates behavioral rationale that explains the interaction between these processes to predict user behavior.
- TRIPLE (CIKM)
  First Author — 2025
  - TRIPLE is a profiling technology that combines the Theory of Planned Behavior (TPB) with Large Language Models (LLMs).
  - Uses LLMs to understand a user's psychological motivations and refines their profile by comparing predictions with actual behavior, dramatically improving personalization services.
- Red Teaming Attack Dataset Analysis
  Project Leader — 2024

- Analyzed real-world LLM vulnerabilities using attack data from the 2023 DEF CON 31 Generative AI Red Teaming Challenge.
- Preprocessed and relabeled the dataset to identify which target categories are most susceptible to harmful-content attacks.
- Classified victim groups and categorized prompt types to reveal systematic patterns in successful jailbreaks.

- Multi-Agent Personality Detection System (PADO)
  Second Author 2024
  - PADO is a multi-agent system that detects personality traits (OCEAN) from user-generated text.
  - Multiple specialized agents collaborate to perform more accurate personality analysis.

- Fashion-FINE
  Third Author 2024
  - Fashion-FINE is a Vision-Language Pre-training model for fine-grained fashion retrieval.
  - Introduces modality-agnostic adapter, hard negative mining with focal loss, and comprehensive cross-modal alignment.

- My Own Style (MOS)
  First Author May. 2022 - 2025
  - Investigated how domain expertise shapes user perception and satisfaction with fashion-recommendation outcomes.
  - Built MOS dashboard that analyzes outfit images and recommends similar/diverse fashion items.
  - Large-scale study with 166 participants showing fashion experts prefer highly similar recommendations while non-experts prefer diverse results.

Publications

Conference Papers

[C.1] Haein Yeo, Seungwan Jin, Taehyung Noh, Yejin Shin, Sangyeon Kang, Sangwoo Heo, Jiwon Chung, Hwarim Hyun, Kyungsik Han (2026). "Can LLMs Persuade Humans with Deception?": From a Deceptive Strategy Taxonomy to a Large-Scale Empirical Study. The ACM Conference on Human Factors in Computing Systems (CHI 2026)

[C.2] Taehyung Noh, Seungwan Jin, Haein Yeo, Kyungsik Han (2026). TRIPLE: Theory-Driven Integration of Planned and Habitual Behaviors for LLM-based Personalization. The AAAI Conference on Artificial Intelligence (AAAI 2026, Oral, Acceptance Rate: 17.6%)

[C.3] Taehyung Noh, Seungwan Jin, Haein Yeo, Kyungsik Han (2025). Externalizing Social-Cognitive Structures for User Modeling: Toward Theory-Driven Profiling with LLMs. The ACM International Conference on Information and Knowledge Management (CIKM 2025 Short, Acceptance Rate: 27.2%)

[C.4] Haein Yeo, Taehyung Noh, Seungwan Jin, Kyungsik Han (2025). PADO: Personality-induced multi-Agents for Detecting OCEAN in human-generated texts. The International Conference on Computational Linguistics (COLING 2025)

[C.5] Seungwan Jin, Hoyoung Choi, Taehyung Noh, Kyungsik Han (2024). Integration of global and local representations for fine-grained cross-modal alignment. The European Conference on Computer Vision (ECCV 2024)

[C.6] Taehyung Noh, Haein Yeo, Myungin Kim, Kyungsik Han (2023). A study on user perception and experience differences in recommendation results by domain expertise: the case of fashion domains. The ACM Conference on Human Factors in Computing Systems Late-Breaking Work (CHI LBW 2023, Acceptance Rate: 36%)

Journal Papers

[J.1] Haein Yeo, Taehyung Noh, Kyungsik Han (2025). LLM-Generated Content-Based Explanations for User Experience in Fashion Recommender Systems. Fashion and Textiles

Technical Reports

[R.1] Yeajin Shin, Kyungsik Han, Taehyung Noh, Mingon Jeong (2025). LLM Attack Strategies. TTA Report

Patents

[P.1] Seungwan Jin, Kyungsik Han, Taehyung Noh, Junghyun Kim (2025). Dynamic Persona-based Prediction System and Method for Personalized Long-term Mental Health Monitoring.

[P.2] Kyungsik Han, Taehyung Noh (2025). Method and Device for Generating User Profiles Using Artificial Intelligence.

[P.3] Bogoan Kim, Dayoung Jung, Taehyung Noh, Kyungsik Han (2023). Visualization Analysis Tool Equipped with VR Multimodal Sensor Data Processing Pipeline Collected from People with Autism Spectrum Disorder.

[P.4] Kyungsik Han, Taehyung Noh, Haein Yeo, Myungjin Kim (2023). Device and Method for Providing Dashboard Services on Fashion Image Analysis.

[P.5] Kyungsik Han, Taehyung Noh, Haein Yeo, Myungjin Kim (2023). Device and Method for Similar Image Recommendation Using Fashion Attributes and Image.

[P.6] Kanghwi Ahn, Junseok Ryu, Kyungsik Han, Taehyung Noh (2023). Method for Analyzing Image and Text-based Characteristics of Objects.

[P.7] Taehyung Noh, Kyungsik Han (2026). Method and Device for Generating User Profiles Using Artificial Intelligence. (PCT International Application, PCT/KR2026/095009)

[P.8] Kyungsik Han, Seungwan Jin, Hoyoung Choi, Taehyung Noh (2022). Image-Text Retrieval Model Construction Method and Service Device.

Software Registrations

[SW.1] Kyungsik Han, Taehyung Noh (2025). Human-in-the-Loop Industrial Preprocessing & Training System (HILIPS). (C-2025-052302)

[SW.2] Kyungsik Han, Taehyung Noh, Haein Yeo, Myungjin Kim (2025). Customized Fashion E-commerce Solutions Dashboard. (C-2025-025185)

[SW.3] Kyungsik Han, Bogoan Kim, Dayoung Jeong, Taehyung Noh (2024). VR Data-based Expert Decision-making and Patient Self-cognition Support Dashboard for Depression Patients. (C-2024-034698)

[SW.4] Kyungsik Han, Taehyung Noh, Haein Yeo, Myungjin Kim (2024). Fashion Element Analysis and Product Description Generation Dashboard for Fashion Experts. (C-2024-016124)

[SW.5] Taehyung Noh, Kyungsik Han (2022). VR User Biometric Signal Data Collection and Analysis System. (C-2022-051352)

Honors and Awards

Best Paper Award, Korea Data Mining Society — Nov. 2024
Best Presentation Award, Korea Software Congress (KSC 2023) — Dec. 2023
Best Inventor Award, Seoul International Invention Fair — Dec. 2022
Participation Award, mySUNI Creative Challenge — Dec. 2022

TEACHING EXPERIENCE

Co-lecturer

- ITE3062: Human-Computer Interaction (In English) — Fall 2024

Teaching Assistant (TA)

- SOI3010: Laboratory practice of Intelligence Computing 2 — Fall 2024

ACADEMIC SERVICES

Conference Reviewer

- Conference on Empirical Methods in Natural Language Processing (EMNLP) — 2024
- ACM Conference on Human Factors in Computing Systems (CHI) — 2024
- ACM International Conference on Information and Knowledge Management (CIKM) — 2023

References

1. Kyungsik Han
   Associate Professor, Hanyang University
   Department of Artificial Intelligence
   kyungsikhan@hanyang.ac.kr